

Alejandro Alvarez Lamazares

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SUMMARY

Computer Science graduate and software engineer currently working as a QA Engineer & DevOps for a production e-commerce platform built with ASP.NET Core.

Specialized in automated testing, backend quality, CI/CD, Linux infrastructure, Docker-based environments and troubleshooting. Designed and developed from scratch a 450-test automated suite covering unit and integration scenarios and integrated it into GitHub Actions to validate Pull Requests before code integration.

Hands-on experience investigating failures across frontend, backend, containers and infrastructure, validating API contracts, authentication, authorization, HTTP error handling, edge cases and idempotency.

In addition to QA and DevOps responsibilities, capable of designing, developing, testing, containerizing and deploying ASP.NET Core/.NET backend applications end-to-end, with experience in PostgreSQL, RabbitMQ, Redis, MongoDB and distributed backend patterns.

EDUCATION

University of Havana — Faculty of Mathematics and Computer Science

Havana, Cuba

Bachelor's Degree in Computer Science, Thesis: "Proposal for an Automated Elective Course Management System Based on a Telegram Bot", Designed a system for managing university elective-course enrollment after gathering requirements directly from university administrative stakeholders., Technologies: Python · Flask · SQLite · python-telegram-bot · Docker · Linux VPS, Conducted multiple requirements discussions with the university secretary., Evaluated technical alternatives according to the university's actual infrastructure and operational constraints., Selected Telegram as the interaction channel to provide an accessible solution for students., Implemented student authentication using initial credentials that users were required to change., Developed functionality allowing students to enroll in elective courses securely through a Telegram bot., Containerized and deployed the solution using Docker and a VPS.

Graduated: July 2025

EXPERIENCE

QA Engineer & DevOps

Jan 2026 – Present

BrincoXpress

Remote

Work on a production e-commerce platform with real users, built around ASP.NET Core and a technical ecosystem including PostgreSQL, MongoDB, Redis, RabbitMQ, Docker and Next.js.

- Designed and developed the project's entire automated testing suite from scratch, growing it from zero tests to approximately 450 automated tests.
- Built approximately 140 unit tests and 310 integration tests, with the complete suite executing in around 3 minutes.
- Implement integration tests using ASP.NET Core WebApplicationFactory to exercise real HTTP endpoints and application behavior.
- Test authentication, authorization, roles, API validation, HTTP status codes, edge cases and idempotent behavior.
- Analyze every new backend functionality, inspect the implementation and independently determine the test cases required to protect existing behavior and cover new requirements.
- Integrated the full regression suite into GitHub Actions, automatically validating Pull Requests before changes are merged.
- Support a development workflow averaging approximately three Pull Requests per week, with automated regression checks applied to incoming changes.
- Produce reproducible technical bug reports that allow developers to identify and reproduce failures efficiently.

- Validate API contracts between backend and frontend, with particular attention to expected HTTP responses, error formats and unhandled edge cases.
- Participate in release validation and provide technical recommendations on whether changes are ready to advance, with final release approval handled by the Project Manager.
- Perform deployments to testing/staging environments using Docker, Docker Compose, Linux VPS, Portainer and Plesk.
- Manage environment variables, application logs, container lifecycle, migrations, reverse proxy configuration and deployment troubleshooting.
- Diagnose failures by correlating container logs, backend behavior, frontend responses and browser developer tools until the root cause is isolated.
- Participate in rollback decisions when API contract changes or backend behavior create incompatibilities with frontend applications.
- Backend Contribution — Background Email Processing: Designed a persistent background processing mechanism to prevent email notifications from being permanently lost when the external email service is temporarily unavailable.
- Store pending tasks in the database instead of relying on synchronous request execution.
- Implemented configurable retry behavior with increasing backoff intervals such as 1 min → 5 min → 15 min → 1 h.
- Made retry limits configurable through environment variables.
- Separate permanently failed tasks from pending work while preserving them for later inspection.
- Enabled failed email tasks to be manually reprocessed from the store administration interface.
- Designed the processing mechanism so its architecture could be extended to additional background-task types, although its current use case is email delivery.

Freelance Software Developer

2024 – 2025

Independent

Remote

Worked directly with real clients to analyze business problems, research technical alternatives and deliver software used in operational environments.

- Order Status Bot (Mar 2024 – Sep 2024): Developed and maintained a Windows automation application for shipment tracking.
- Built a Python-based tool that retrieved shipment status information from multiple logistics providers and automatically updated structured Excel workbooks.
- Implemented web automation and scraping using technologies including Selenium, BeautifulSoup, Requests and Playwright.
- Manipulated Excel files programmatically using openpyxl.
- Implemented logic to prevent completed shipments from being unnecessarily processed again.
- Added support for additional logistics providers as client requirements evolved, including integrations for carriers such as TIPSA, Correos Express and DHL.
- Delivered the application in a form that allowed the end user to launch it directly from Windows without interacting with Python or source code.
- Continued maintaining and expanding the solution in response to new client requests.
- WhatsApp Customer Service Automation (2025): Designed and deployed a customer-service automation solution used in production for several months.
- Researched and implemented an alternative WhatsApp integration architecture using WAHA and n8n when direct access to Meta for Developers was not available.
- Built automated conversational workflows capable of answering questions about business processes, commercial procedures and technical product specifications.
- Structured product and offer information in PostgreSQL so automated agents could query business information effectively.
- Added support for processing customer voice messages.
- Integrated WhatsApp communication through the WAHA API.
- Deployed the solution using Docker on a Linux VPS.
- Continuously expanded the system during 2025 as new requirements emerged from meetings with business stakeholders.

- The system operated with real customers for several months before the company moved customer service back to a human-operated model.

PROJECTS

OrderFlow API | .NET Backend Engineering Portfolio Project (2026 – Present). REST API developed as a long-term backend engineering project intended to demonstrate end-to-end software engineering and system-design capabilities.

- Technologies and concepts: C# · ASP.NET Core · PostgreSQL · Docker · RabbitMQ · Microservices · DDD · CQRS · Outbox Pattern · Automated Testing · CI/CD
- Design and implementation of backend APIs using ASP.NET Core.
- Domain modeling using Domain-Driven Design principles.
- PostgreSQL persistence and database migrations.
- Containerized development and execution with Docker.
- Asynchronous communication using RabbitMQ.
- Exploration and implementation of microservice boundaries and distributed-system patterns.
- Implementation of CQRS and transactional messaging patterns such as Outbox.
- Automated testing as a core part of the development process.
- CI/CD workflows for automated validation.
- Developed incrementally as a practical application of advanced backend concepts and architectural decision-making.

TECHNICAL SKILLS

Backend: C# · .NET · ASP.NET Core · REST APIs · Entity Framework Core · Python · Flask

Testing & Quality Engineering: xUnit · Moq · WebApplicationFactory · Unit Testing · Integration Testing · End-to-End Testing · API Testing · Regression Testing · Authentication & Authorization Testing · Idempotency Testing · HTTP Contract Validation

DevOps & Infrastructure: Docker · Docker Compose · GitHub Actions · CI/CD · Linux · VPS Administration · Plesk · Portainer · Git · Environment Configuration · Application Logs · Reverse Proxy · Deployment Troubleshooting

Data & Messaging: PostgreSQL · MongoDB · Redis · RabbitMQ · SQLite

Additional Tools: Playwright · Selenium · BeautifulSoup · Requests · openpyxl · n8n · WAHA

CERTIFICATIONS

EF SET Certificate, 78/100, Oct 2025

LANGUAGES

Spanish: Native English: C2 Proficient — EF SET Certificate, 78/100, Oct 2025

PROFESSIONAL INTERESTS

QA Automation · Software Testing · DevOps · Backend Engineering · Distributed Systems · CI/CD · Software Reliability